

National Academy of Science and Technology
(Affiliated to Pokhara University)
Dhangadhi, Kailali
Accredited by University Grants Commission (UGC), Nepal 2022

Pre-University Examination

Level: Bachelor
Program: B.E. Computer
Course: **Computer Architecture**

Semester: V_Fall

Year : 2022
F.M : 100
P.M : 45
Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate between Computer Architecture and Computer Organization. Draw the diagram of extended IAS computer. 8
b) Define RTL. Explain all micro operation with RTL code. 7
2. a) What is normalization in floating point operation? Explain how to store floating point numbers in a computer memory. 8
b) Perform -7×6 using Booth's algorithm. 7
3. a) Differentiate between hardwired and microprogrammed control unit. 7
b) In certain scientific computation it is necessary to perform the arithmetic operation $(A_i + B_i)/C_i$ with a stream of numbers. Specify a pipeline configuration to carry out this task. List the contents of all registers in the pipeline for $i = 1$ to $i = 6$. 8
4. a) Why mapping is needed in cache? Explain three different cache mapping schemes. 8
b) Explain different RAID systems. 7
5. a) Differentiate between programmed I/O and interrupt driven I/O. How DMA is better than programmed I/O and interrupt driven I/O? 7

- b) Instruction pipelining increases system performance without increasing processor number, explain how? What are the major hurdles in pipelining and how they are overcome? 8
6. a) Describe Flynn's classification. 8
- b) Explain different types of hardware and software performance issues in multicore computers. 7
7. Write short notes on any two: 2×5
- a) *Dual core and quad core processor*
 - b) *RISC vs CISC*
 - c) *Cache Coherence*

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Pre-University Examination

Level: Bachelor

Semester: V_Fall

Year : 2022

Program: B.E. Computer

F.M : 100

Course: Probability and Statistics

P.M : 45

Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

- 1.a) An article in Environment Concentration and Toxicology report the amount of cadmium in sea scallops observed at a number of different station in North Atlantic water. The observed value are as follows:

5.1, 14.1, 14.7, 10.8, 6.5, 7.7, 14.1, 9.5, 3.7, 8.9, 7.9, 7.9, 4.5,
10.1, 5.0, 9.6, 5.5, 5.1, 11.4, 8.0, 12.1, 7.5, 8.5, 13.1, 6.4, 18.0,
27.0, 18.9, 10.8, 13.1, 8.4, 16.9, 2.7, 9.6, 4.5, 12.4, 5.5, 12.7,
17.1, 20.5, 22.3

- i) Construct a frequency distribution for the data set using 0.0 –4.0 and 4.0-8.0 as the second class interval respectively.
ii) Draw the histogram and frequency curve of the above frequency distribution.
- b) An analysis of monthly wages paid to the workers on two firms A and b belonging to the same industry gives the following results:

[7]

	Firm A	Firm B
Number of worker:	450	500
Average monthly wages in \$	180	175
Variance of distribution of wage:	81	100

- i) Which firm, A or B, has a larger wage bill ?
ii) In which firm, A or B is there greater variability in wage ?
iii) Calculate combined mean and combined variance of wage of firm A and firm B.

[8]

2. a) The contents of urns I, II and III are as follows:

1 white, 2 black and 3 red balls

2 white balls, 1 black and 1 red balls.

4 white, 5 black and 3 red balls.

One urn is chosen at random and two balls are drawn. They happen to be white and red. What is the probability that they come from urns I, II or III?

[7]

OR

State and prove Baye's theorem.

b) The weights (in grams) of 10 orange picked at random from a basket are as follows: 45, 30, 75, 40, 60, 65, 70, 85, 80, 90

i. List the five number summaries.

ii. Form the box and whisker plot and describe the shape.

[8]

3. a) The number of accident in a year attributed to taxi driver in a city is Poisson distribution with mean 3. Out of 1000 taxi driver, find approximately the number of driver with

i) more than 3 accidents in a year

ii) less than 2 accident

[7]

b) In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and S.D. of the distribution.

[8]

4. a) Joint probability distribution of random variables X and Y is given by

$$F(x,y) = \begin{cases} k(6-x-y), & 0 < x < 2, 2 < y < 4 \\ 0 & \text{otherwise} \end{cases}$$

Find:

i) the constant k

ii) Marginal density function of X and Y

iii) $E(X)$ and $V(X)$

iv) Are X and Y independent?

[8]

b) A population consists of value 7, 6, 8, 4, 10. Prove that sample mean of size 3 is unbiased estimator of the population mean.

[7]

5.a) In a simple of 600 high school students from a city, 400 are found to use dot pens. In one of 900 from a neighboring city, 450 are found to use dot pens. Do the data indicate that the cities are significantly different with respect to the habit of using dot pens around the students? ($\alpha=5\%$) [7]

b) Eleven college students were given a test in statistics. They are given a month's tuition and a second test was held at the end of it. Do the marks give evidence that the student have benefited by the extra coaching. (use $\alpha=0.05$) [8]

Students	1	2	3	4	5	6	7	8	9	10	11	12
Marks in 1 st test	23	20	19	21	18	20	18	17	23	16	19	24
Marks in 2 nd test	24	19	22	18	20	22	20	20	23	20	18	22

6. a) Suppose that the life time of a certain kind of computer is random variable x having the Gamma distribution with parameter $\alpha=2$ and $\beta=3$, if the daily capacity of this city is 9 million gallon of water, what is the probability that on any given day the water supply is inadequate? [7]

b) The following table gives the age X and blood pressure Y of 10 Women

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	149	150

i) Find correlation coefficient between age and pressure.

ii) Fit a regression equation that best describe the above data. [8]

7. Write short notes on. (Any two)

[5x2=10]

a) Criteria of good estimator

b) Conditional Probability and marginal probability

c) Characteristics of Normal Distribution.

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Pre-University Examination

Level: Bachelor
Program: B.E. Computer
Course: Numerical Methods

Semester: V_Fall

Year : 2022

F.M : 100

P.M : 45

Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Find a root of the equation: $x \log_{10} x = 12.34$ using any root bracketing method correct to three decimal places. Give a reason for the choice of the method. [8]

- b) Using any non bracketing method, find a root of the equation $x^3 - 5x + 1 = 0$ correct to four decimal places. [7]

2. a) Estimate the missing value in the following data using suitable interpolation formula. [7]

x	0	1	2	3	4
y	3	13	28	-	78

- b) The train resistance R (lbs/ton) is measured for the following values of velocities V (km/hr): [8]

V	20	40	60	80	100
R	14	20	24	26	27

If R and V have relation of the form $R = ae^{bv}$, find the values of a and b using least square approximation. Also, estimate R at $V = 50$ km/hr from the fitted curve.

3. a) Solve the following system of equations using a factorization method. [8]

$$x + y + z = 1;$$

$$4x + 3y - z = 6;$$

$$3x + 5y + 3z = 4.$$

- b) Solve the following system of equations using Gauss-Seidel method. [7]

$$x + 6y - 10z = -3; \quad 10x - 5y - 2z = 3; \quad 4x - 10y + 3z = -3.$$

4. a) Evaluate the integral $\int_0^1 \frac{1}{1+x^2} dx$ using Romberg method. [8]

b) Using three point Gaussian integration, evaluate $\int_0^1 \frac{1}{\sqrt{1+x^4}} dx$ [7]

5. a) Using R-K 4th order method, solve $\frac{dy}{dx} = -2xy^2$ with $y(0)=1$ in the interval $0 \leq x \leq 0.5$ taking step size $h = 0.25$. [7]

b) Using Heun's method, solve $\frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$ with $y(0) = 1$ and $y'(0) = 0$. Estimate $y(1)$ and $y'(1)$ taking step size $h = 0.5$. [8]

6. a) Solve the equation $u_{xx} + u_{yy} = 1+x+y$ over the square domain $0 \leq x \leq 1$ and $0 \leq y \leq 1$ with $u=0$ on the boundary and mesh length $h=1$. [8]

b) Find the largest eigen value and corresponding eigen vector of the matrix using power method. [7]

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

7. Attempt any two

[2 X 5 = 10]

- Write and classify the general form of second order partial differential equation.
- Solve $y' = x+y$, $y(0)=1$ using Picard's method up to second iteration.
- Find the dy/dx at $x=1$ using the data:

x	1	1.2	1.4	1.6
y	9	9.5	10.2	11.0

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Pre-University Examination

Level: Bachelor
Program: B.E. Computer
Course: Operating System

Semester: V_Fall

Year : 2022
F.M : 100
P.M : 45
Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Define Operating System. Explain with different types with example. 8
b) Do you think a process can exist without any state? Justify your view with the help of process state transition diagram and PCB. 7

2. a) What is memory management? Explain memory hierarchy. 5
b) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling algorithms. 10
 - i. FCFS
 - ii. SJF
 - iii. Priority (Preemptive)
 - iv. HRRN

Process	Arrival Time	Burst Time	Priority
A	0	3	5
B	2	6	4
C	4	4	1
D	6	5	3
E	8	2	2
F	3	4	1

3. a) Given five memory partitions of 100 KB, 500KB,200KB,300KB 7

and 600KB (in order), how would the first-fit, best fit and worst-fit algorithms place processes of 212KB, 417KB, 112KB, and 426KB (in order)? Which algorithm makes the most efficient use of memory? Illustrate.

8

- b) Suppose a disk drive has 5500 cylinders, numbered 0 to 5499. The drive is currently serving a request at cylinder 2243 and the previous request was at cylinder 1125. The queue of pending requests in FIFO order is 586, 1470, 1913, 1774, 5348, 1509, 5022, 1750, 130. Starting from the current head position what is the total distance that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms?

- i. FCFS
- ii. Closest Cylinder Next//SSTF
- iii. C-Scan (Initially moving upward)
- iv. Scan (Initially moving downward)

4. a) What is the difference between absolute and relative path name of a file? What criteria should be used to decide which strategy? (Contiguous, linked, indexed, allocation) is best utilized for a particular file?

7

- b) Define Deadlock. Explain the different techniques used to prevent from deadlock.

8

5. a) Compare and contrast between virus and worm. Explain Access Control List (ACL) and Access Control Matrix (ACM).

7

- b) Explain different classical IPC problems in brief.

8

6. a) Define Distributed System. List out advantages and disadvantages of distributed system over centralized system.

7

- b) Write the difference between logical and physical address. Describe about Coalescing and Compaction with suitable examples.

8

7. Write short notes on: (Any two)

2×5

- a) The Shell

- b) Types of Kernels

- c) Windows 2000

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Pre-University Examination

Level: Bachelor

Program: B.E. Computer

Course: Theory of Computation

Semester: V_Fall

Year : 2022

F.M : 100

P.M : 45

Time : 3Hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Define FA and regular expression. Construct a finite automata that accepts the set of string that neither has '11' or '00' as substring over $\Sigma = \{0, 1\}$.

7

- a) Consider the following NFA with ϵ -transition.

8

δ/Σ	a	b	c	ϵ
$\rightarrow q_0$	$\{q_0\}$	$\{\Phi\}$	$\{\Phi\}$	$\{q_1\}$
q_1	$\{\Phi\}$	$\{q_1\}$	$\{\Phi\}$	$\{q_2\}$
$*q_2$	$\{\Phi\}$	$\{\Phi\}$	$\{q_2\}$	$\{\Phi\}$

Convert this NFA with ϵ -transition into NFA without ϵ -transition.

2. a) When the grammar is ambiguous? For given grammar rule:

$S \rightarrow aSa$

$S \rightarrow bSb$

$S \rightarrow c$

Design PDA for the given grammar.

- b) Define CNF. Convert the following grammar into CNF.

$S \rightarrow abSb/a/aAb$

$A \rightarrow bS/aAAb$

8

7

3. a) Give the formal definition of pushdown automata. Construct a PDA accepting the language $L = \{a^n b^m c^n \mid n > 0, m > 0\}$

7

- b) State pumping lemma. Prove that language $L = \{a^n b^{2n} \mid n > 0\}$ is not regular language.

8

- 4.a) Why we prefer PDA rather than FA? State the closure properties of CFL's. 8
- b) Prove that the following function is computable: $f(n): n + 1$. 7
- 5.a) Describe about Universal Turing machine. What is the application of Turing Machine? 7
- b) What is recursive and recursively enumerable language? Show that the union of two recursive language is also recursive. 8
- 6.a) What do you understand by time and space complexity? Explain. 7
- b) Explain NP-complete problems with suitable example. 8
7. Write short notes on following (**Any Two**) 5x2
- a) Relation and Function
 - b) Church Turing Thesis
 - c) Halting Problem

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Year : 2022

Program: B.E. Computer

F.M : 100

Course: **Computer Graphics**

P.M : 45

Time : 3Hrs

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) What do you understand about computer graphics? Mention some of advantages of computer graphics 7
b) Explain architecture of Raster scan system with importance of video controller. 8
2. a) Derive Bresenham's line drawing algorithm slope greater than one. 7
b) Derive midpoint Ellipse algorithm of region 1. 8
3. a) Explain 2D transformation using a homogeneous coordinate system. 8
b) What is clipping? Explain Cohen Sutherland's line clipping algorithm. 7
4. a) Derive quadratic cubic bezier curve. and explain the Bezier curve properties. 8
b) Explain and Derive orthographic and oblique parallel projection transformation matrix. 7
5. a) What is the hidden surface? Explain the Z-buffer method. 8
b) What is ambient light? Derive illumination model. 7
6. a) Explain Gouraud and Phong shading methods with advantages and disadvantages. 8
b) Explain about GKS and different kinds of file formats. 7
7. Write short notes on **any two**: 2×5
 - a) Color model and its type.
 - a) Back Face detection.
 - Fractal Geometry Method